



Nowcasting Inflation:

From Theory to a Working Model with Cambodia as Evidence

A joint CAPRED-GDDE initiative for data-driven economic policy

PART 1

Concepts

What nowcasting is, the theory behind it, and the one equation everything rests on

01

What is inflation nowcasting?

DEFINITION

Estimating inflation for a month that has already happened or is happening now before the statistics office publishes it.

Now + forecast: the number already exists in the economy. We just cannot see it yet.

-  **Backcast** a month that has ended but is not yet published
-  **Nowcast** the month we are living in
-  **Forecast** months that have not started.

August 2026, as it happened



the nowcasting window

Why bother?

Because we want to know what we can expect in the upcoming week especially during the economic shock.

A nowcast is a conditional expectation

$$\hat{\pi}_t = E[\pi_t \mid \Omega_t]$$

π_t

Inflation this month: the monthly log change in the CPI.
Unknown until NIS publishes.

Ω_t

The information set: everything observable today, fuel
decreases to date, past CPI, food price etc.

$E[\cdot \mid \cdot]$

The conditional expectation: of all possible guesses, the
one with the smallest mean squared error.

Inflation is a weighted average of price changes

$$\pi_t = \Delta \log(\text{CPI}_t) = \sum_i w_i \cdot \Delta \log(p_{i,t})$$

$$w_{\text{fuel}} \cdot \Delta \log p_{\text{fuel},t}$$

Observed every day; the decreed pump price

Fuels are 5.2% of the basket

+

$$\sum_{i \neq \text{fuel}} w_i \cdot \Delta \log p_{i,t}$$

Unobserved until NIS publishes

Food 44.8% · housing 17.1%

Two channels carry fuel into the rest

- **Direct:** a 10% pump-price rise adds 0.52% by arithmetic alone.
- **Indirect:** fuel is a cost of moving food and running services.

Why fuel is the indicator to watch

5.2%

of the CPI basket

fuels, COICOP 07.2.2 (weight 5.175)

27%

**of month-to-month headline
volatility**

0 days

publication lag

decreed prices are known the day they change



Fuel: the price is decreed

The government sets pump prices. The price we record is exactly the price people pay. No sampling error.



Food: the price is surveyed

Price is an averaged price from many outlets and days, so its food index is smoother than daily market prices.

Deriving Statistical Model from Inflation Definition

1

Start from the index

$$\Delta \log \text{CPI} = w_{\text{fuel}} \cdot \Delta \log \text{Fuel} + \sum_{i \neq \text{fuel}} w_i \cdot \Delta \log p_{i,t}$$

2

Summarise the others

$$\sum_{i \neq \text{fuel}} w_i \cdot \Delta \log p_{i,t} = \alpha + \gamma \cdot \Delta \log \text{Fuel} + \varepsilon$$

3

Collect the terms

$$\Delta \log \text{CPI} = \alpha + (w_{\text{fuel}} + \gamma) \cdot \Delta \log \text{Fuel} + \varepsilon$$

Step 2, line by line

1 Project each non-fuel item's price change on fuel

$$\Delta \log p_{i,t} = a_i + g_i \cdot \Delta \log \text{Fuel}_t + u_{i,t}$$

2 Multiply each item by its basket weight

$$w_i \cdot \Delta \log p_{i,t} = w_i a_i + w_i g_i \cdot \Delta \log \text{Fuel}_t + w_i u_{i,t}$$

3 Add up every non-fuel item

$$\sum w_i \cdot \Delta \log p_{i,t} = \sum w_i a_i + (\sum w_i g_i) \cdot \Delta \log \text{Fuel}_t + \sum w_i u_{i,t}$$

4 Name the three sums

$$\alpha = \sum w_i a_i \quad \gamma = \sum w_i g_i \quad \varepsilon_t = \sum w_i u_{i,t}$$

Each $u_{i,t}$ is orthogonal to fuel by construction, so ε has mean zero and averages across dozens of items.

One equation carries the whole method

$$\Delta \log (\text{CPI}_t) = \alpha + \beta \cdot \Delta \log (\text{Fuel}_t) + \varepsilon_t$$

$\Delta \log \text{CPI}_t$

Target

Month-on-month inflation. Unknown until NIS publishes.

α

Drift

Average month-on-month inflation from everything that is not fuel.

β

Pass-through

Percent change in the CPI for a 1% change in fuel

$\Delta \log \text{Fuel}_t$

Indicator

This month's change in the decreed pump price. Known daily.

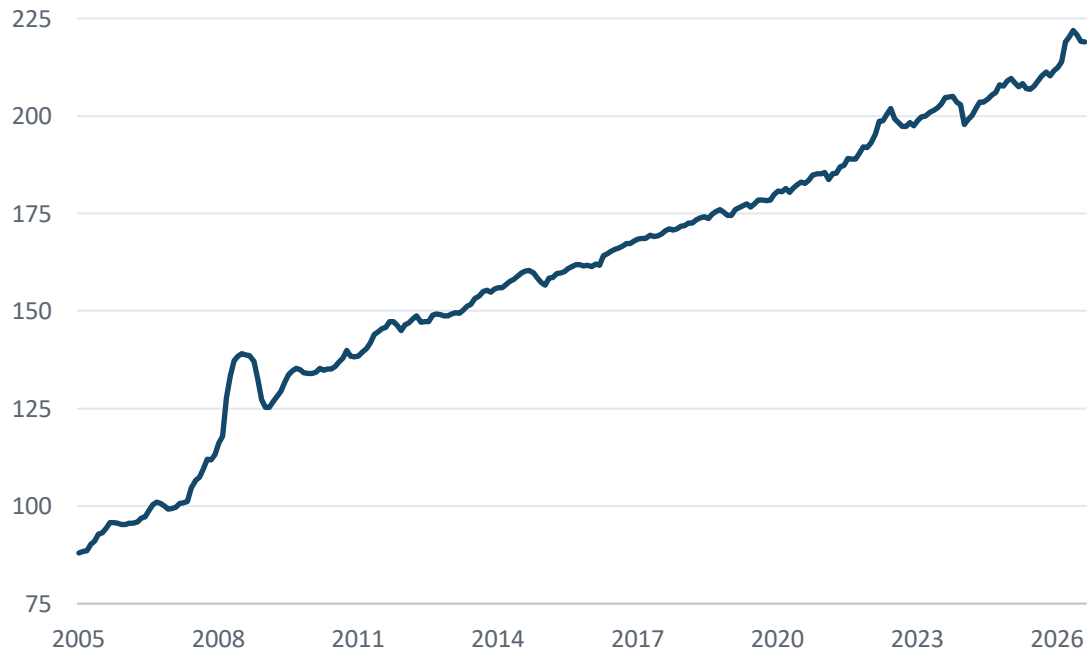
ε_t

Error term/unobserved

Mean zero.

Why changes, not levels: levels trend, changes do not

CPI level, Phnom Penh (November 2006 = 100)

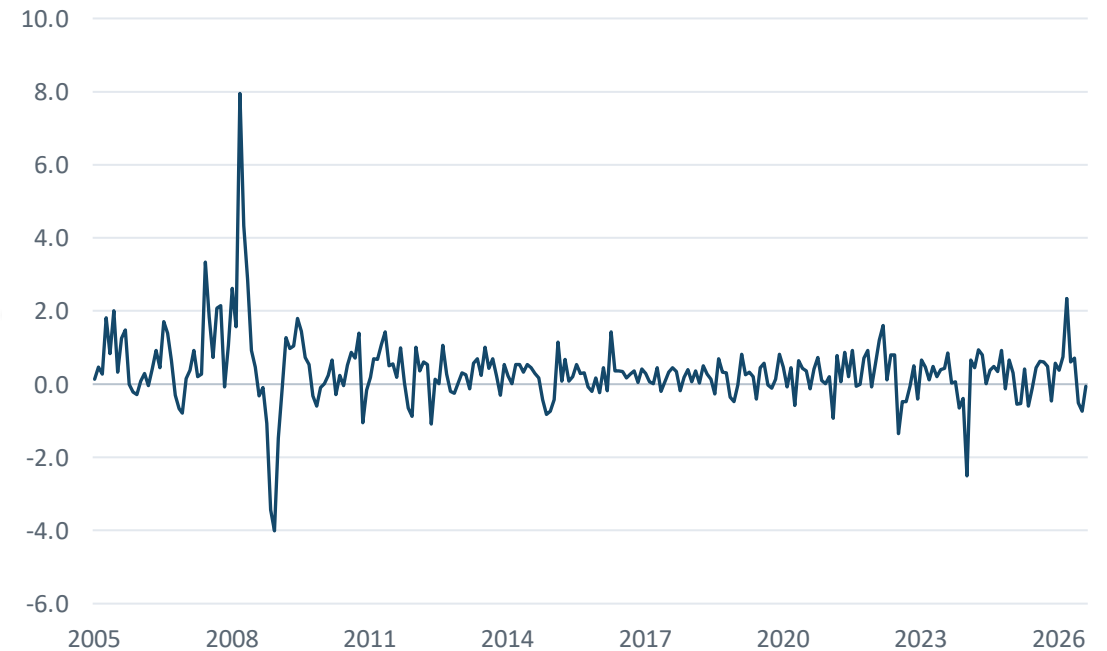


Trending: ADF test $p = 0.31$

A p-value of 0.31 in an Augmented Dickey-Fuller (ADF) test means this time series data is non-stationary. Therefore, we need to difference them.

$$\Delta \log(\text{CPI}_t) = \log(\text{CPI}_t) - \log(\text{CPI}_{t-1})$$

Monthly log change of the CPI, %



Mean-reverting: ADF test $p < 0.001$

Stationarity is a requirement for SARIMA and other ML algorithms. Mean and variance are constant over time.

The Random Walk Forecast is the Bar Every Model Must Clear

THE RANDOM WALK FORECAST

$$\hat{\pi}_t = \pi_{t-1}$$

This month's inflation equals last month's.

No model, no parameters, no data beyond the target's own last value.

Also known as a naïve forecast, RW is a standard benchmark for time-series forecasting

1

It is the free forecast

A model costs data, code and judgement. It has to earn that cost.

2

Beating it proves new information

If extra data and parameters cannot beat 'same as last month' out of sample, they add noise, not knowledge.

3

It is hard to beat

Atkeson & Ohanian (2001): US inflation models failed to beat a naive forecast.

The scorecard

Relative RMSE = $\text{RMSE}(\text{model}) \div \text{RMSE}(\text{random walk})$. Below 1.00 beats it.

Twelve readings average 2.5%, so the rule forecasts 2.5%

Month	Inflation	Month	Inflation
Month 1	2.1%	Month 7	2.7%
Month 2	2.8%	Month 8	2.4%
Month 3	2.5%	Month 9	2.6%
Month 4	1.9%	Month 10	2.5%
Month 5	2.2%	Month 11	2.3%
Month 6	3.0%	Month 12	3.0%

PAST 12 MONTHS

2.5%

the rule carries it forward

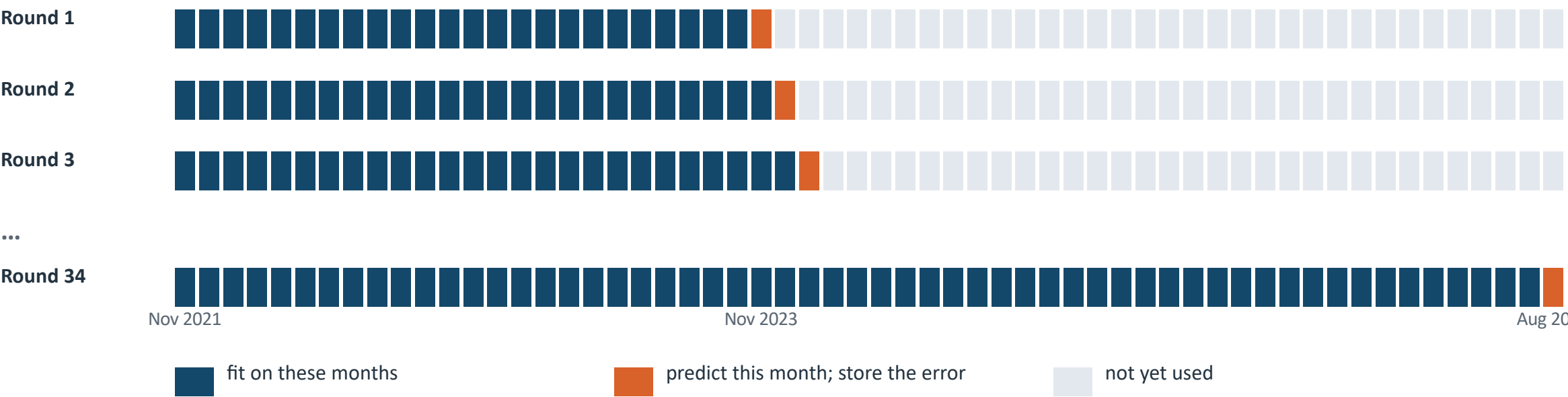
NEXT 12 MONTHS, FORECAST

2.5%

Nothing else enters: no unemployment gap, no oil price, no model.

$\pi_{t-12 \rightarrow t} = 2.5\%$ so $\hat{\pi}_{t \rightarrow t+12} = 2.5\%$ this is the number every model has to beat

Back Testing Over 34 Months



58

months with both CPI and fuel
Nov 2021 – Aug 2026

24

months for the first fit
the window then grows by one

34

months scored
each predicted by a model that never saw it

02

PART 2

The four models

Random walk · SARIMA · Bridge Regression · Ridge

Four models, from no information to the most

	MODEL	EQUATION	WHAT IT CAN SEE	PARAMETERS	ROLE
1	Random walk	$\hat{\pi}_t = \pi_{t-1}$	Last month's inflation only	0	Benchmark
2	SARIMA	$(1 - \varphi L)(1 - \Phi L^{12})(1 - L)p_t = c + (1 + \theta L)\varepsilon_t$	Its own past and seasonal pattern	4	Baseline
3	Bridge OLS	$\pi_t = \alpha + \beta \cdot f_t + \varepsilon_t$	This month's fuel price change	2	The workhorse
4	Ridge	$\min \Sigma (\pi_t - \alpha - \beta z_t)^2 + \lambda \beta^2$	Same as bridge, slope shrunk	$2 + \lambda$	Ready for many predictors

π_t month-on-month inflation · f_t monthly log change in fuel · z_t standardized f_t · L the lag operator, $L\pi_t = \pi_{t-1}$ · p_t log CPI, so $(1 - L)p_t = \pi_t$

Random walk: this month equals last month

$$\hat{\pi}_t = \pi_{t-1}$$

How it works

- No estimation: carry the latest published month forward.
- Implicitly assumes every shock to inflation is permanent.
- In a live nowcast, “last month” is the newest NIS figure.



Strengths

- Zero parameters, so zero estimation error
- Tracks momentum when inflation is persistent
- The yardstick every model is scored against



Weaknesses

- Copies last month's noise into this month
- Always one step late at turning points
- Ignores everything observed since

0.96 pp

RMSE over 34 months

1.00

vs random walk, by definition

−0.74%

August 2026 call; actual −0.06%

In our data

Month-on-month inflation here is barely persistent (autocorrelation 0.23, 2021–26), so copying last month copies noise. Predicting zero (0% month-on-month inflation) does better: 0.80 pp.

SARIMA: inflation explained by its own past

$$(1 - \phi L)(1 - \Phi L^{12})(1 - L) p_t = c + (1 + \theta L) \varepsilon_t$$

How it works

- p_t is log CPI; $(1 - L)$ takes the monthly difference, turning the trending level into inflation
- ϕ persistence · Φ the same month last year · θ last month's surprise · c the drift
- Fitted by maximum likelihood on the differenced series; refit every month



Strengths

- Needs no outside data
- Captures persistence, seasonality and the trend



Weaknesses

- Cannot see a fuel decree, or any outside shock
- Short samples strain it: 12-month lags on 24–57 months

0.82 pp

RMSE over 34 months

0.85

vs random walk

−0.04%

August 2026 call; actual
−0.06%

In our data

Unbiased, and 15% better than the random walk on the headline, but still behind predicting zero (0.82 vs 0.80 pp).

Bridge regression: the foundation equation at work

$$\pi_t = \alpha + \beta \cdot f_t + \varepsilon_t$$

How it works

- Average the daily decreed price to a monthly level; take its log change f_t
- Fit α and β by least squares on months where CPI and fuel are both known
- Plug in this month's observed f_t — the bridge from daily data to a monthly target



Strengths

- Uses information from the month being nowcast
- Two parameters, each with an economic meaning



Weaknesses

- Sees only what fuel drives, blind to food (44.8%)
- Passes every fuel move through in full, at once

0.69 pp

RMSE over 34 months

0.71

vs random walk

+0.60%

August 2026 call; actual
-0.06%

Ridge: the bridge with a brake on the slope

$$\min \sum_t (\pi_t - \alpha - \beta z_t)^2 + \lambda \beta^2$$

How it works

- z_t is the fuel change standardized to mean = 0 and s.d. = 1
- The penalty $\lambda \beta^2$ pulls β toward zero: a little bias, less variance
- One predictor: $\beta = \sum z_t(\pi_t - \bar{\pi}) \div (\sum z_t^2 + \lambda)$; we use $\lambda = 1$



Strengths

- Stable when predictors are many and correlated
- A small, controlled departure from OLS
- Hoerl & Kennard (1970): well understood



Weaknesses

- With one predictor it changes almost nothing
- λ must be chosen on out-of-sample RMSE

0.68 pp

RMSE over 34 months

0.71

vs random walk

+0.60%

August 2026 call; actual
-0.06%

In our data

With 57 months, $\sum z^2 = 57$, so ridge keeps $57 \div 58 = 98\%$ of the OLS slope and matches the bridge. It is there for the day we add online price indices by class.

PART 3

Our work as evidence

Cambodia, November 2021 – August 2026: estimates, scores, and the August miss

03

What we measure, and with what

Outcome

NIS Consumer Price Index, Phnom Penh, all items and 12 divisions, monthly.
We model its as monthly log change.

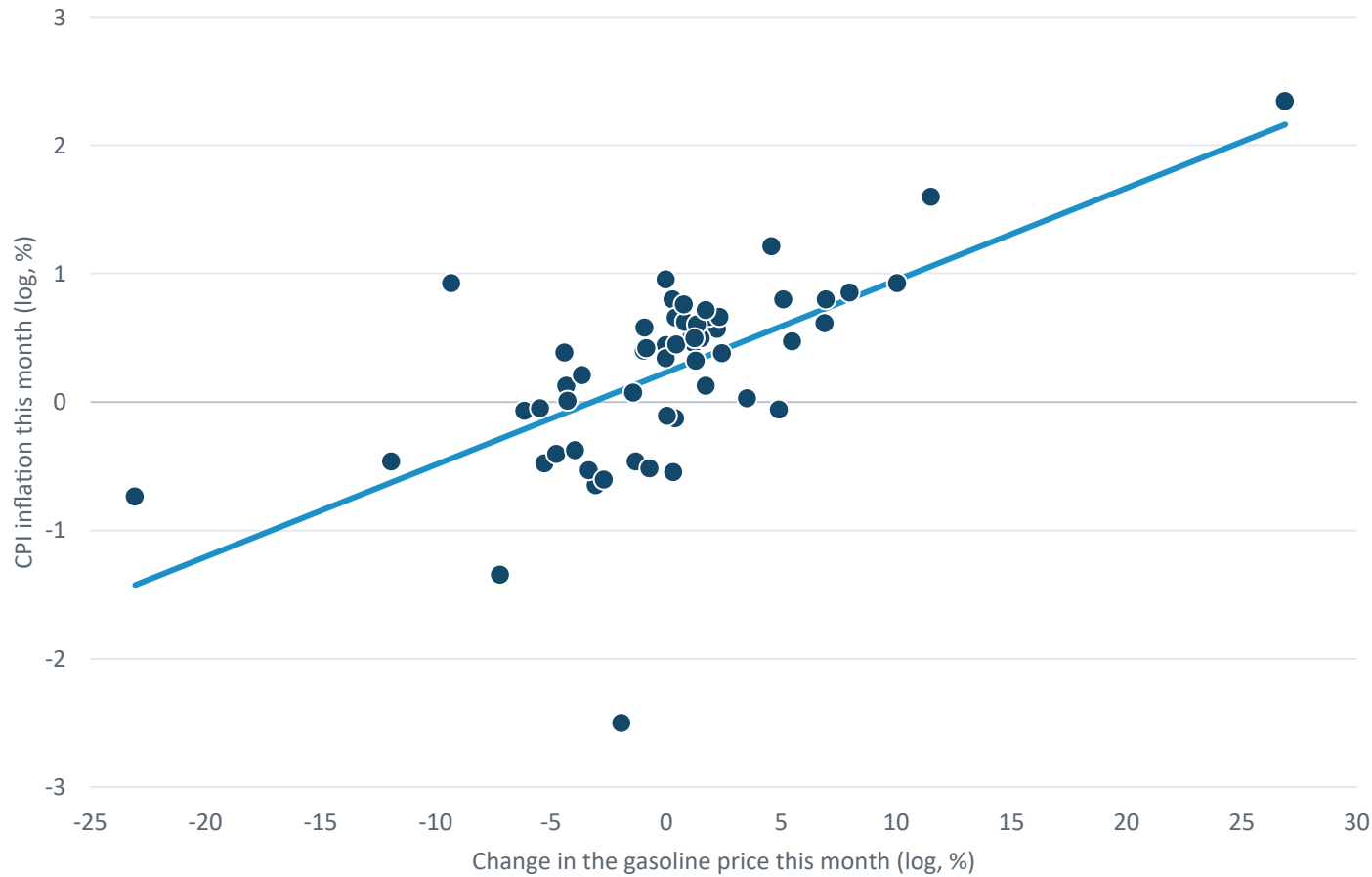
Independent variable

Decreed pump price of regular gasoline, daily, averaged to a monthly level.
Diesel, LPG and lags were tested and not selected.

Sample

58 months with both series (Nov 2021 – Aug 2026): 24 to start the first fit, 34 scored out of sample.

The foundation equation, estimated



$$\beta = 0.072$$

$t = 6.2$ · a 10% fuel rise adds 0.72% to the CPI

$$\alpha = 0.23\% \text{ a month}$$

the drift from everything that is not fuel

$$R^2 = 0.41$$

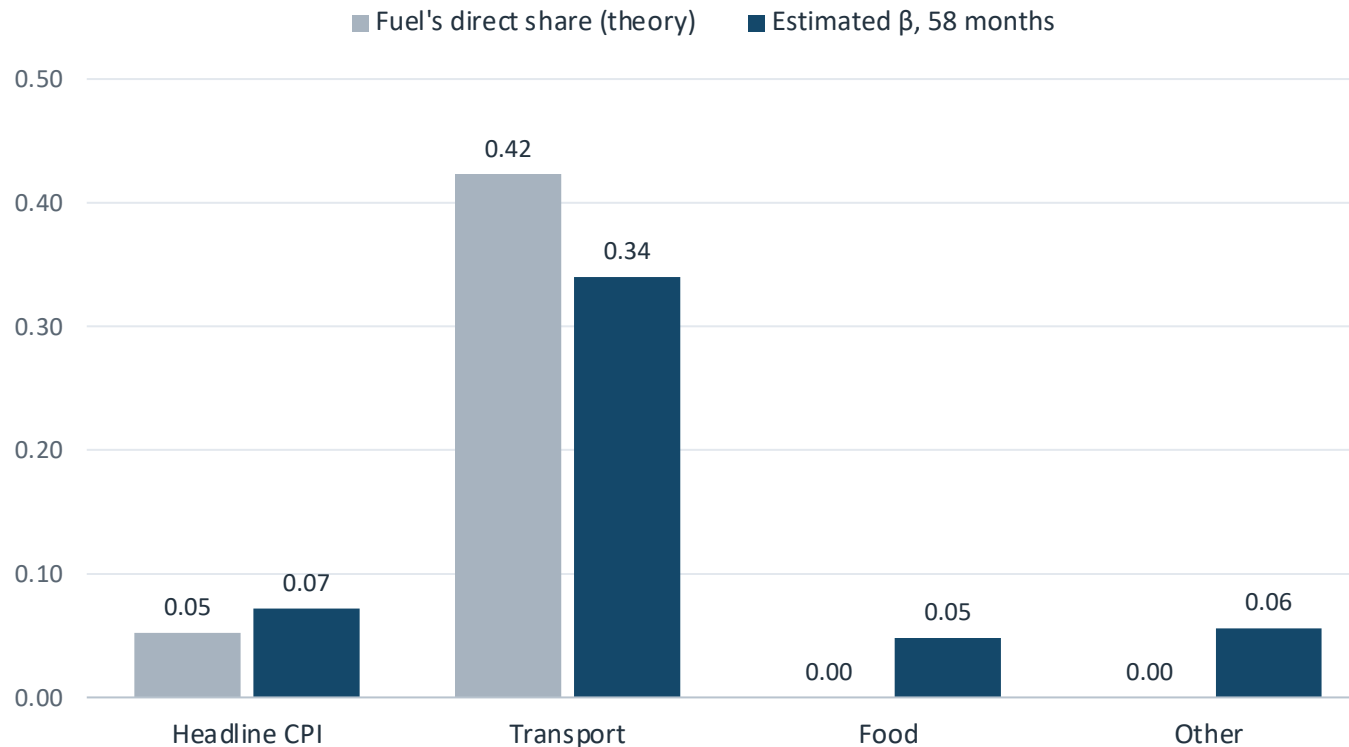
fuel explains 41% of monthly variation; the rest is ϵ

$$n = 58 \text{ months}$$

November 2021 – August 2026, full sample

Does the data agree with the theory?

Fuel's direct share of each index vs the estimated β



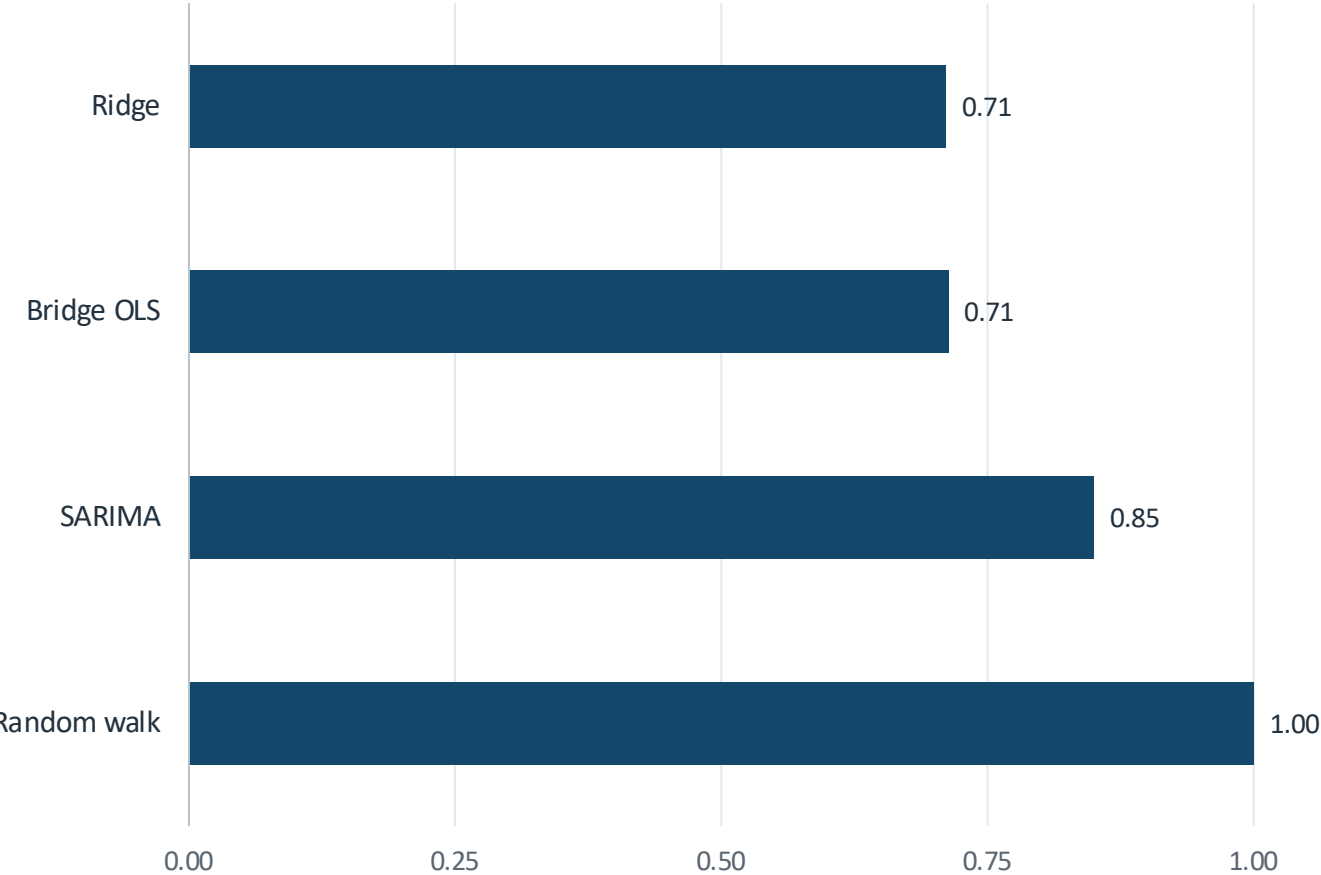
- **Headline: β above the direct share**
0.072 against 0.052 (base-period weight); fuel's direct effect today is nearer 0.03, so other effects add about 0.04.
- **Transport: β close to the share**
0.34 against 0.42: fuel's real weight is now about 0.38 (NIS), and our pump-price measure doesn't exactly match the price NIS records.
- **Food: no fuel in the basket, yet $\beta > 0$**
 $\beta = 0.048$, $t = 3.6$; pure indirect pass-through, via transport and distribution costs.
- **Other: no pump fuel, yet $\beta > 0$**
 $\beta = 0.056$, $t = 5.45$; mostly housing and utilities (including cooking gas) and restaurants.

The closer to the pump, the tighter the fit. R^2 : Transport 0.76 · Headline 0.41 · Food 0.19 · Other 0.35

Weighted back up: with price-updated weights, food 0.56×0.048 + transport 0.075×0.32 (whole-month β) + other $0.37 \times 0.056 = 0.072$, exactly the headline's β .

Scores: all beat the random walk, none decisively

RMSE relative to the random walk, headline CPI, 34 months (lower is better)



29% lower error

bridge vs random walk: 0.69 against 0.96 pp

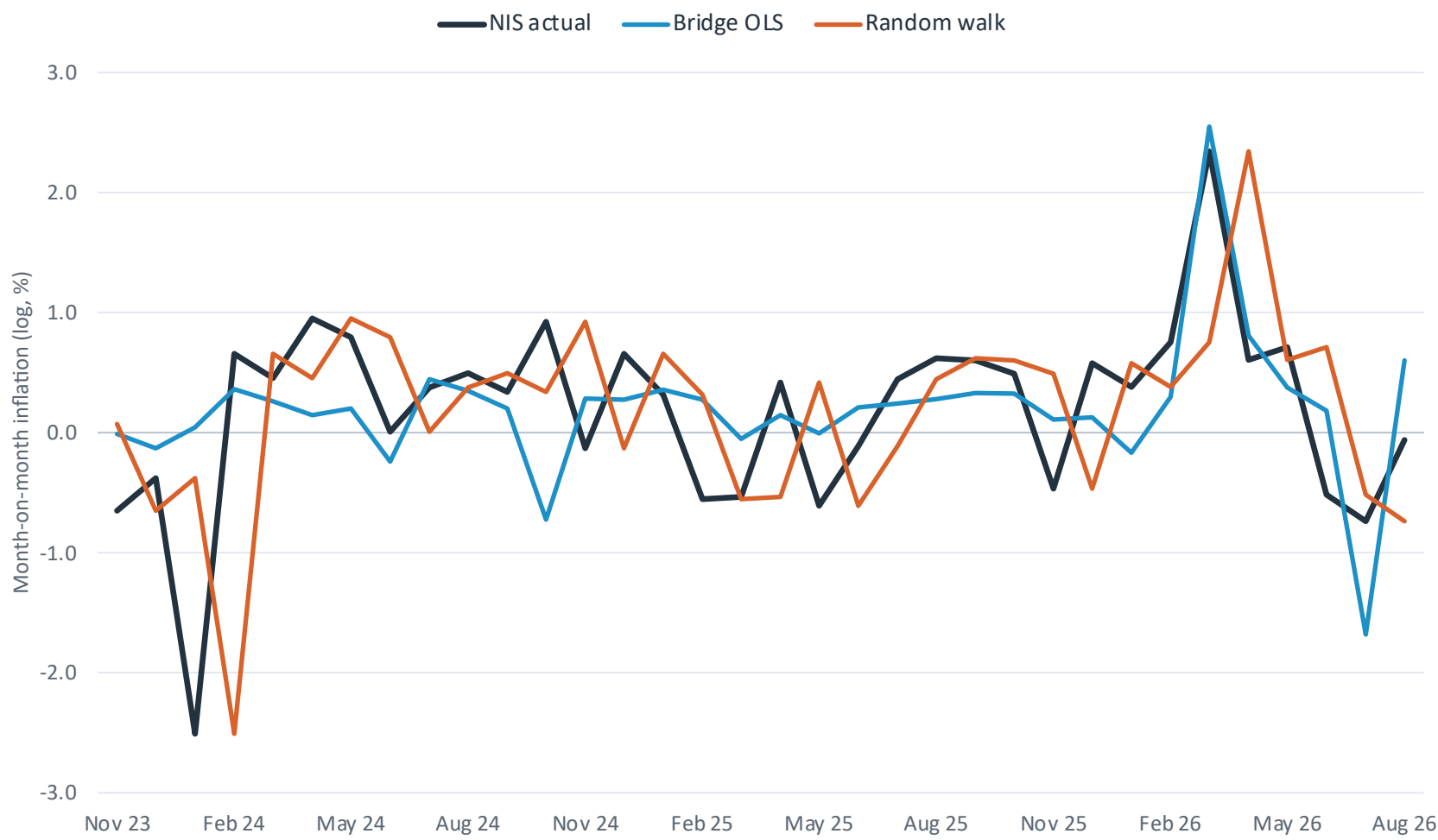
Sample still too small

Diebold–Mariano test against the random walk, 34 months: $p = 0.17$ to 0.26 across the three models. Keep scoring every month until $N = 80$ months

Bridge, by target

Transport 0.37 · Headline 0.71 · Food 0.79 (vs random walk)

Month by month: the bridge follows fuel, the random walk lags



Random walk: one month late

It repeats last month's number, so every surprise reappears a month later.

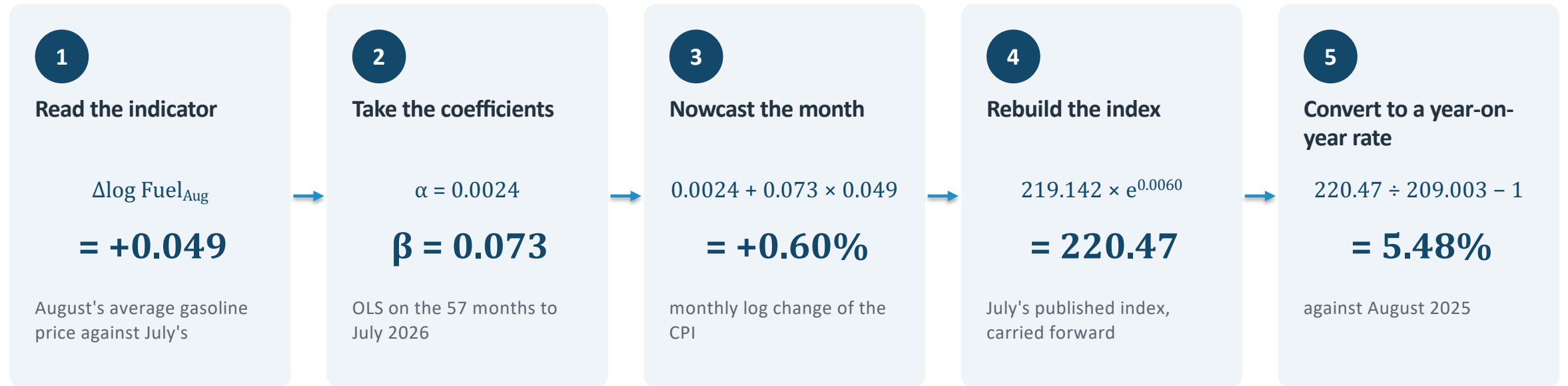
Bridge: moves with fuel

Right direction in big fuel months; too big when pass-through stalls. July 2026: -1.68% against -0.74%.

RMSE, 34 months

Bridge 0.69 pp · random walk 0.96 pp

Worked example: the August 2026 nowcast, step by step



OUR NOWCAST

5.48%

NIS PUBLISHED

4.79%

ERROR

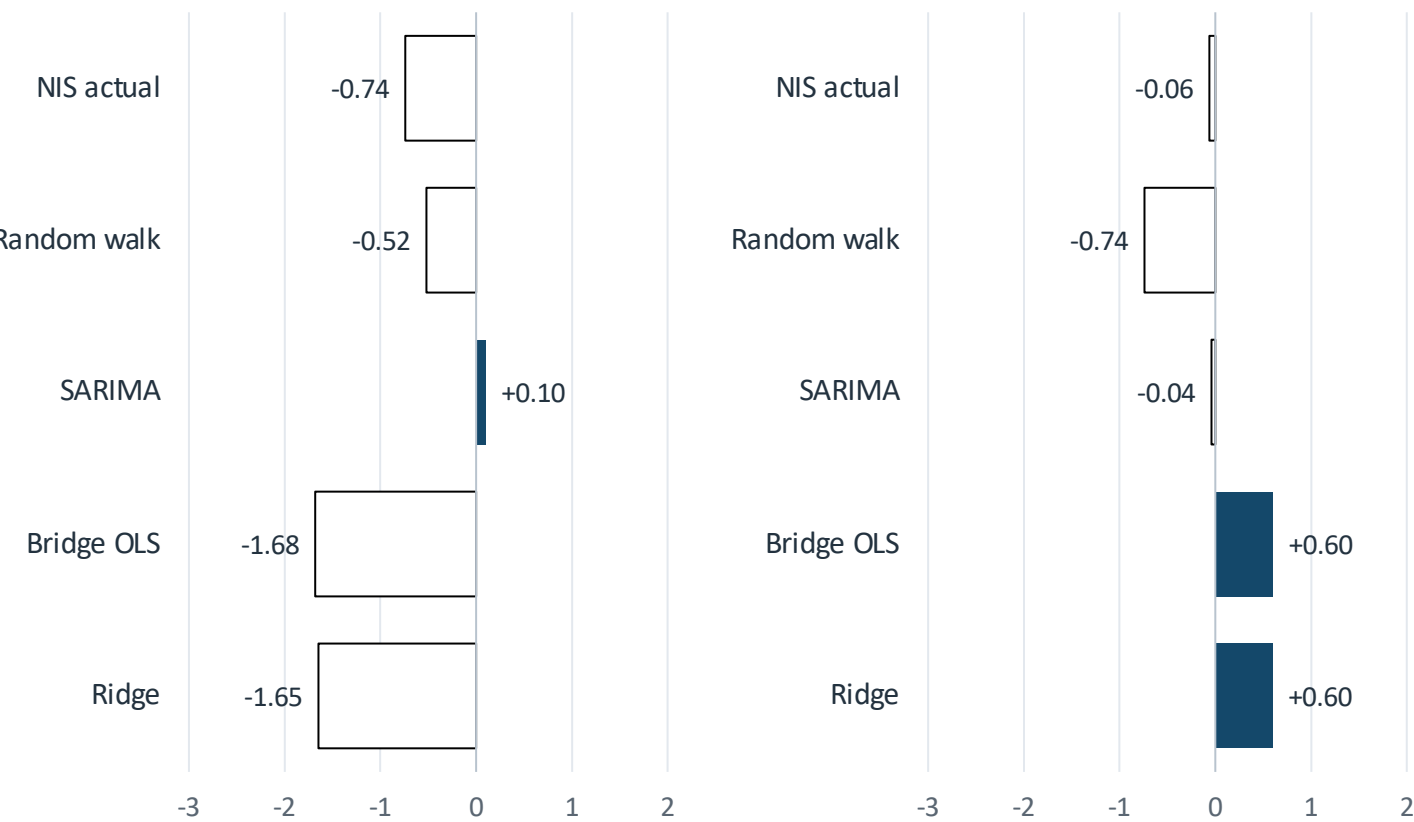
+0.69 pp

July and August 2026: four models against the truth

Month-on-Month inflation, predicted and actual (log, %)

July 2026 · gasoline -20.6%

August 2026 · gasoline +5.0%



July: gasoline fell 21%

Bridge and ridge said -1.68% and -1.65%: right direction, more than twice NIS's -0.74%. The random walk, repeating June, was closest at -0.52%.

August: gasoline rose 5%

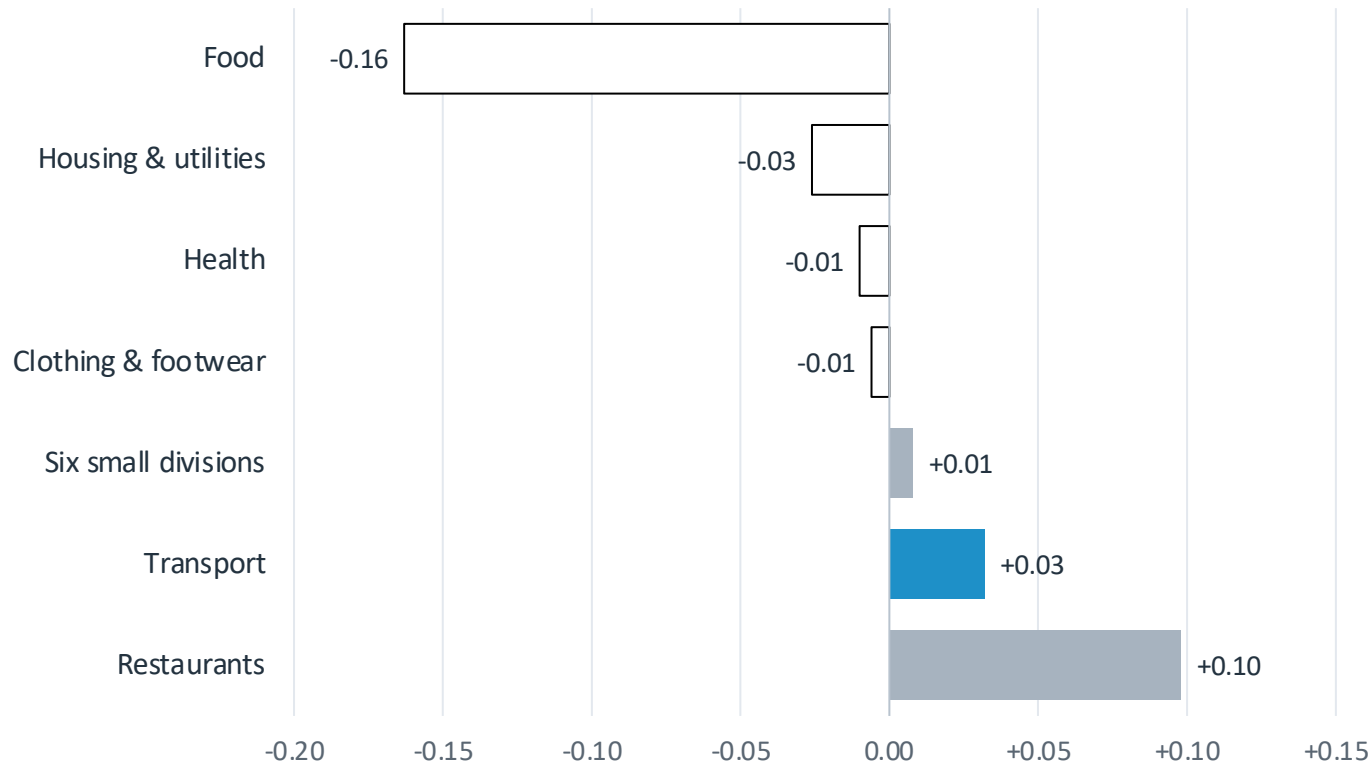
Bridge and ridge said +0.60% against NIS's -0.06%. SARIMA was closest at -0.04%; the random walk, repeating July, said -0.74%.

Two months do not rank models

RMSE over July and August: random walk 0.50 pp, SARIMA 0.59, bridge 0.81. Over all 34 months the order reverses: bridge 0.69, SARIMA 0.82, random walk 0.96.

Why August missed: food fell while fuel rose

Contribution to August's headline change, pp (weight × division change)



Sum of divisions -0.07 pp • NIS headline -0.06 pp

Each bar is the division's weight times its own log change. Food: $0.448 \times (-0.36\%) = -0.16$ pp. Restaurants: $0.059 \times (+1.70\%) = +0.10$ pp. Transport: $0.122 \times (+0.26\%) = +0.03$ pp. The twelve contributions sum to -0.07 pp against the published -0.06 pp; the 0.01 pp gap is rounding of the weights.



The model's only input: fuel +4.9%

The bridge turned it into +0.60% for the whole CPI.



Transport behaved

Its own equation implied +0.46%; it rose only by +0.26% (NIS).



Food did not

Its own equation implied +0.55%; it fell -0.36% on 44.8% of the basket.



Not a broken equation

Fitted on 2026 alone, $R^2 = 0.73$, the tightest yet. Splitting the CPI into 12 divisions was tested and lost.

Four things our own data have shown

1

The theory holds

β tracks fuel's share of each index, and the weighted divisions reproduce the headline to within 0.01 pp.

2

Fuel beats the random walk

About 29% lower error over 34 out-of-sample months — real, but not yet statistically decisive.

3

Logs and changes are not optional

The CPI level trends (ADF $p = 0.31$); its log change does not. Big fuel moves are symmetric only in logs.

4

The August miss came from food

Food fell while fuel rose. The equation was fine; nothing in it can see food.

Next: add food price as a second regressor. We can also add the online price index.

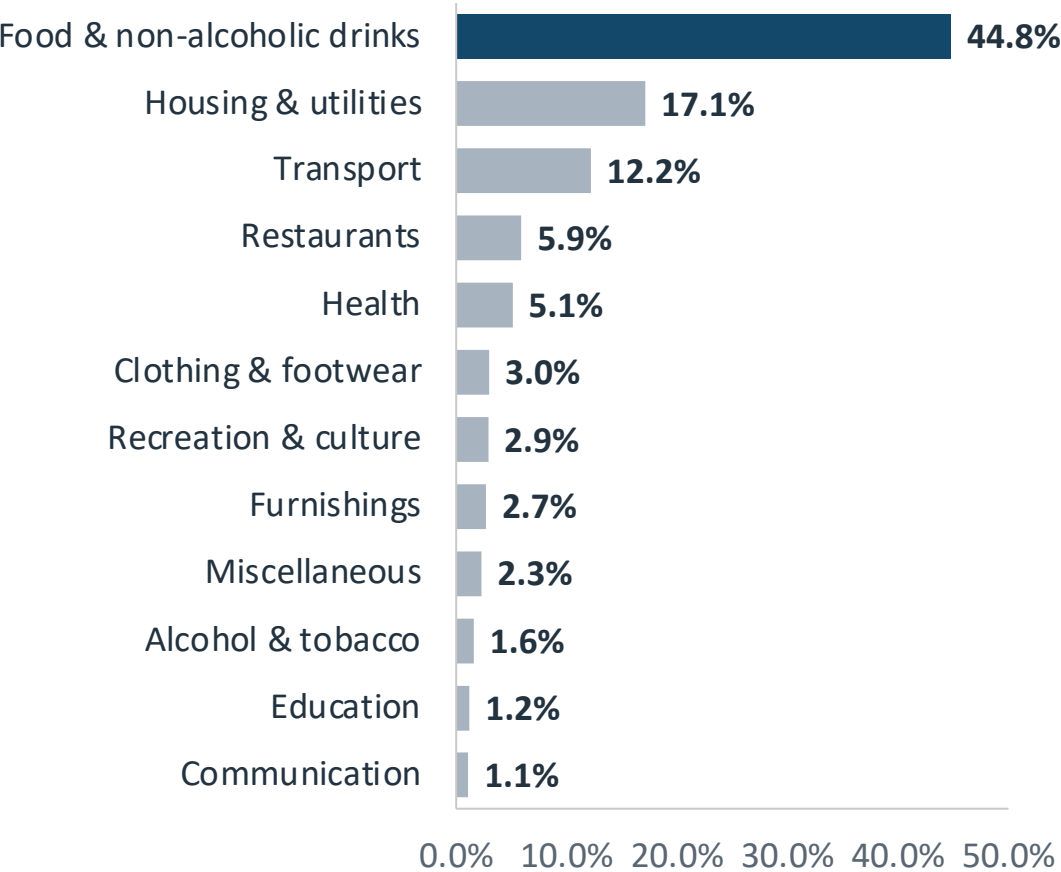
Food is the only block that is big, volatile and visible early

Test for a new predictor	Food	Housing & utilities	Alcohol & tobacco
Share of the basket	44.8%	17.1%	1.6%
Monthly movement	Volatile	Sticky: rents and decreed tariffs	Volatile
Visible before NIS publishes	Scraped daily	No daily source	Scraped daily

Weight times movement is what lands in ε. Food is 45% of the basket and reprices every month, so it dominates the error we are trying to remove. Housing is 17% and mostly sticky rents and decreed tariffs, so there is little monthly variation left to predict.

We do scrape alcohol & tobacco. At 1.6% of the basket, even a 5% price jump moves the headline by 0.08 pp, below the error either model makes. Its value is validation: check the scraper against the published NIS class while the food history accumulates.

What the CPI basket weighs (NIS weights, %)



Thank You

AI & Big Data for Economic Monitoring in Cambodia

A CAPRED initiative • Supported by the Australian Government